

Pointer trace example

```
struct Quidditch {  
    int quaffle;  
    int *snitch;  
    int bludger[2];  
};
```

```
struct Hogwarts {  
    int wizard;  
    Quidditch harry;  
    Quidditch *potter;  
};
```

Pointer trace example

```
Quidditch * hufflepuff(Hogwarts * cedric) {
    Quidditch *seeker = &(cedric->harry);
    seeker->snitch = new int;
    *(seeker->snitch) = 2;
    cedric = new Hogwarts;
    cedric->harry.quaffle = 6;
    cedric->potter = seeker;
    cedric->potter->quaffle = 8;
    cedric->potter->snitch =
        &(cedric->potter->bludger[1]);
    seeker->bludger[0] = 4;
    return seeker;
}

void gryffindor() {
    Hogwarts *triwizard = new
Hogwarts[3];
    triwizard[1].wizard = 3;
    triwizard[1].potter = NULL;
    triwizard[0] = triwizard[1];
    triwizard[2].potter =
hufflepuff(triwizard);
    triwizard[2].potter->quaffle = 4;
}
```

Pointer trace example

```
Hogwarts *triwizard = new Hogwarts[3];
```

Stack (gryffindor):

Hogwarts *
triwizard

Heap:

int wizard = ?	int wizard = ?	int wizard = ?
Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?
Quidditch * potter = ?	Quidditch * potter = ?	Quidditch * potter = ?

Pointer trace example

```
triwizard[1].wizard = 3;
```

Stack (gryffindor):

Hogwarts *
triwizard



Heap:

int wizard = ?	int wizard = 3	int wizard = ?
Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?
Quidditch * potter = ?	Quidditch * potter = ?	Quidditch * potter = ?

Pointer trace example

```
triwizard[1].potter = NULL;
```

Stack (gryffindor):

Hogwarts *
triwizard



Heap:

int wizard = ?	int wizard = 3	int wizard = ?
Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?
Quidditch * potter = ?	Quidditch * potter = NULL	Quidditch * potter = ?

Pointer trace example

```
triwizard[0] = triwizard[1];
```

Stack (gryffindor): Heap:

Hogwarts *
triwizard



int wizard = 3	int wizard = 3	int wizard = ?
Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?
Quidditch *potter = NULL	Quidditch *potter = NULL	Quidditch *potter = ?

Pointer trace example

```
triwizard[2].potter = hufflepuff(triwizard);
```

Stack (gryffindor):

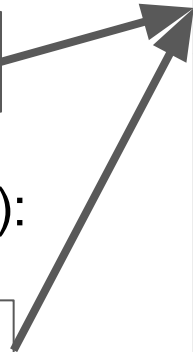
Hogwarts *
triwizard

Heap:

int wizard = 3	int wizard = 3	int wizard = ?
Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?	Quidditch harry int quaffle = ? int *snitch = ? int bludger ? ?
Quidditch * potter = NULL	Quidditch * potter = NULL	Quidditch * potter = ?

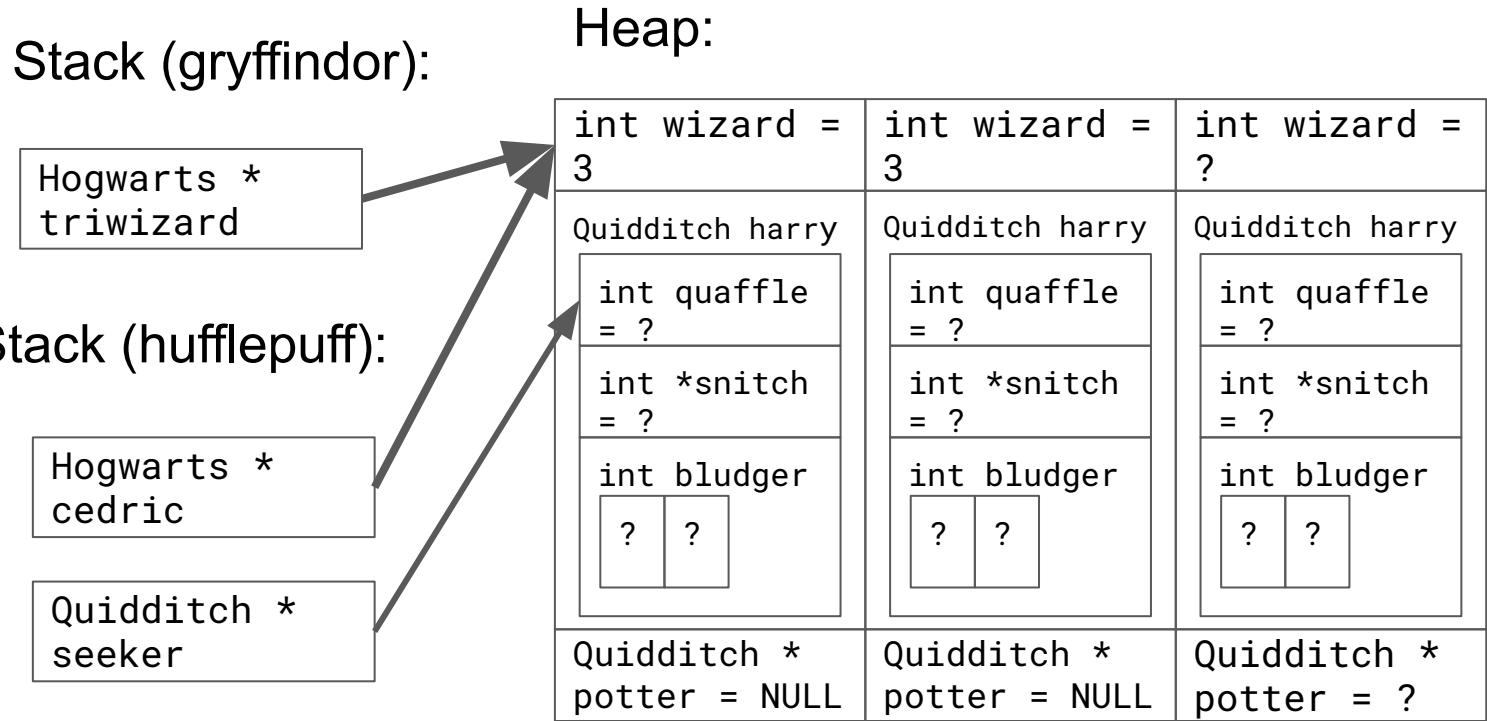
Stack (hufflepuff):

Hogwarts *
cedric



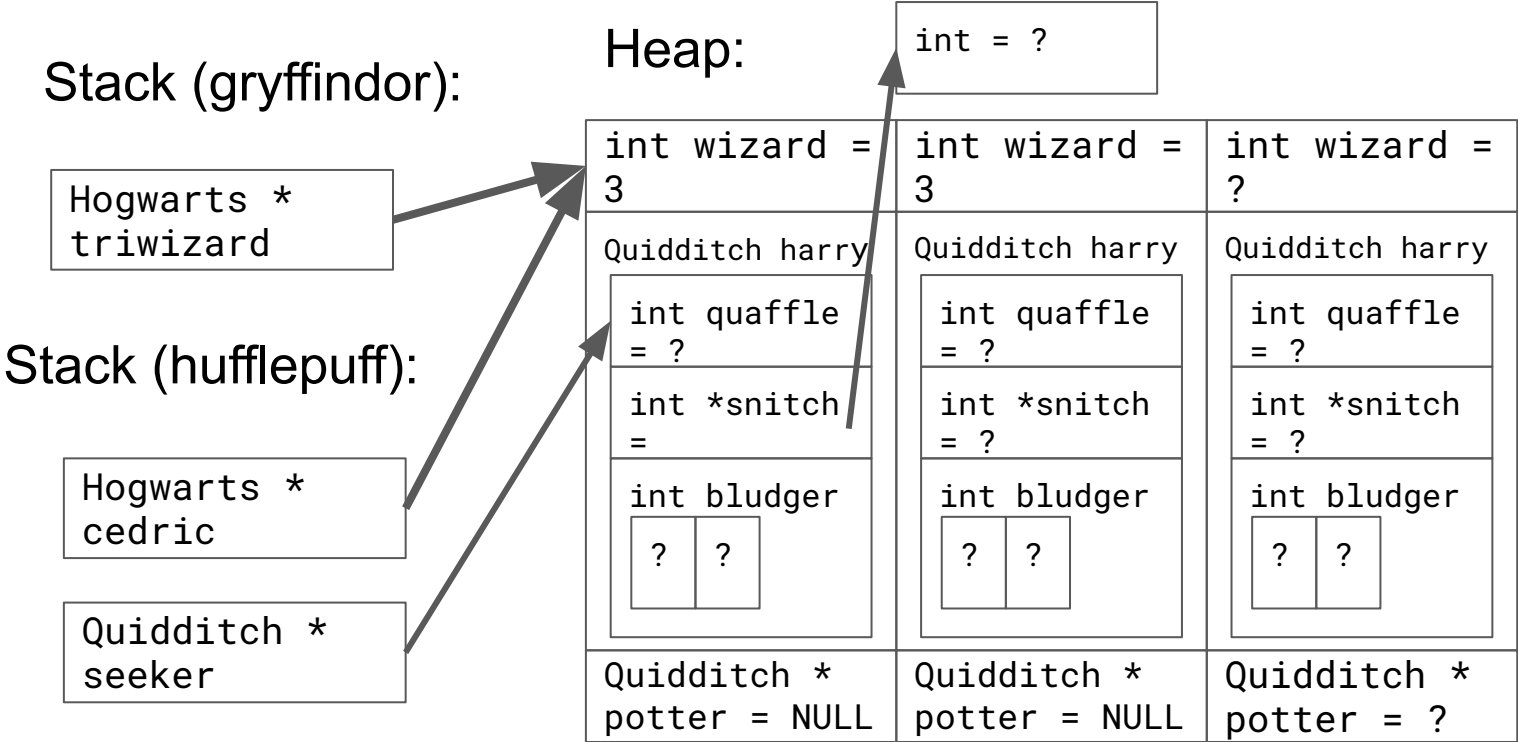
Pointer trace example

```
Quidditch *seeker = &(cedric->harry);
```



Pointer trace example

```
seeker->snitch = new int;
```



Pointer trace example

```
*(seeker->snitch) = 2;
```

Stack (gryffindor):

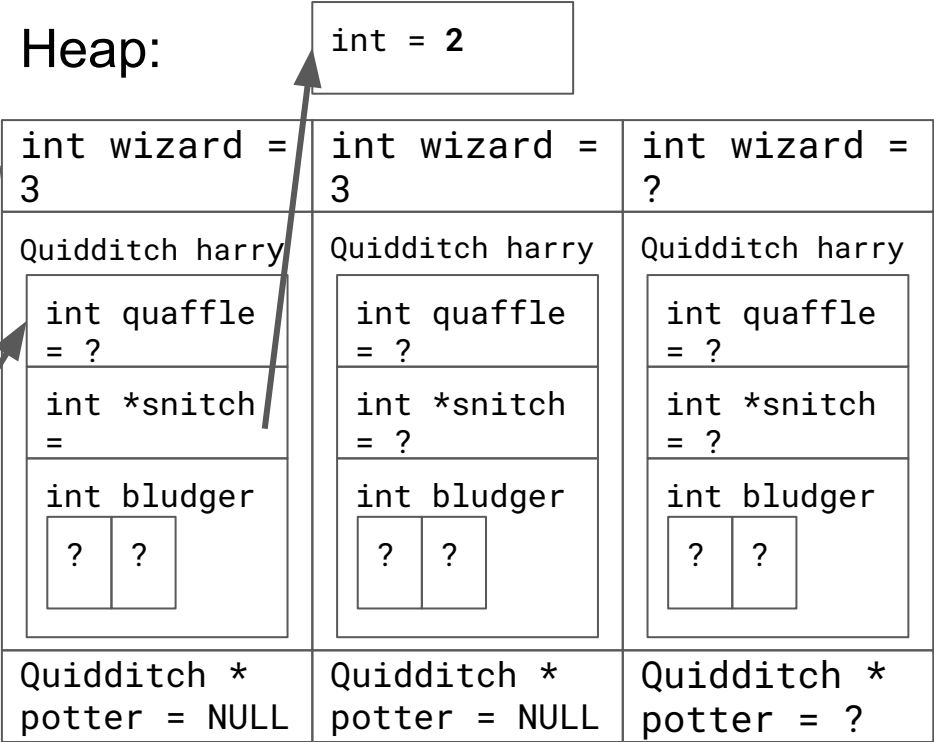
Hogwarts *
triwizard

Stack (hufflepuff):

Hogwarts *
cedric

Quidditch *
seeker

Heap:



Pointer trace example

```
cedric = new Hogwarts;
```

Stack (gryffindor):

```
Hogwarts *  
triwizard
```

Heap:

```
int wizard =  
?
```

Quidditch harry

```
int quaffle  
= ?
```

```
int *snitch  
= ?
```

```
int bludger
```

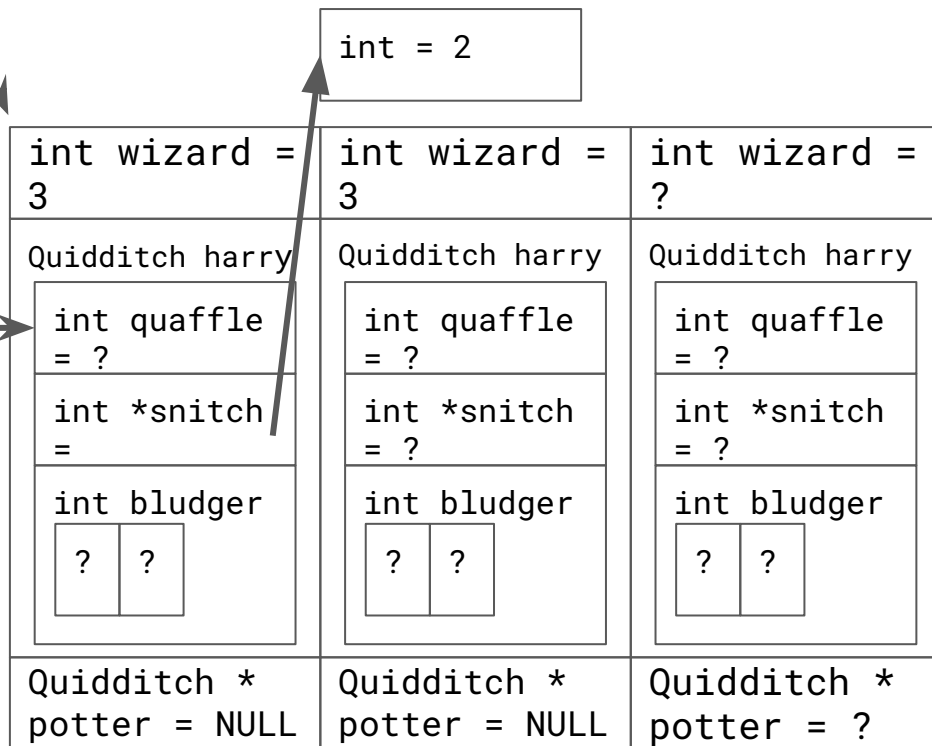
```
? ?
```

```
Quidditch *  
potter = ?
```

Stack (hufflepuff):

```
Hogwarts *  
cedric
```

```
Quidditch *  
seeker
```



Pointer trace example

```
cedric->harry.quaffle = 6;
```

Stack (gryffindor):

Hogwarts *
triwizard

Heap:

```
int wizard =  
?
```

Quidditch harry

```
int quaffle
= 6
```

```
int *snitch
= ?
```

```
int bludger
```

?	?
---	---

Quidditch *
potter = ?

Stack (hufflepuff):

```
Hogwarts *
cedric
```

Quidditch *
seeker

```
int = 2
```

```
int wizard =
3
```

Quidditch harry

```
int quaffle
= ?
```

```
int *snitch
=
```

```
int bludger
```

?	?
---	---

```
Quidditch *
potter = NULL
```

```
int wizard =
3
```

Quidditch harry

```
int quaffle
= ?
```

```
int *snitch
= ?
```

```
int bludger
```

?	?
---	---

```
Quidditch *
potter = NULL
```

```
int wizard =  
?
```

Quidditch harry

```
int quaffle
= ?
```

```
int *snitch
= ?
```

```
int bludger
```

?	?
---	---

Quidditch *
potter = ?

Pointer trace example

```
cedric->potter = seeker;
```

Stack (gryffindor):

Hogwarts *
triwizard

Heap:

int wizard =
?

Quidditch harry

int quaffle
= 6

int *snitch
= ?

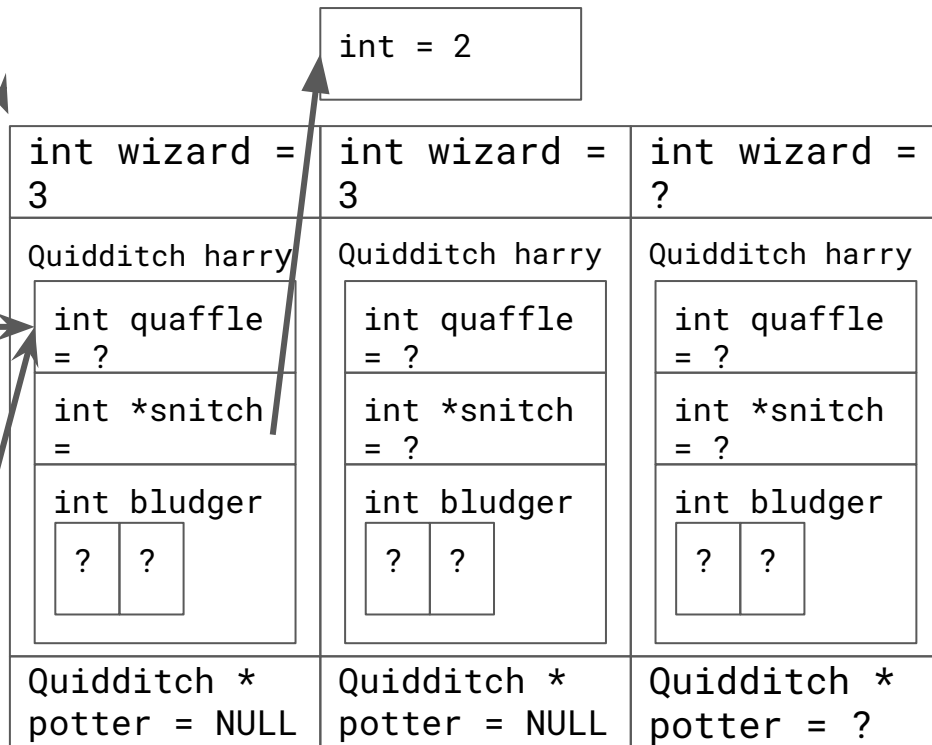
int bludger
? ?

Quidditch *
potter =

Stack (hufflepuff):

Hogwarts *
cedric

Quidditch *
seeker



Pointer trace example

cedric->potter->quaffle = 8;

Stack (gryffindor):

Hogwarts *
triwizard

Heap:

int wizard =
?

Quidditch harry

int quaffle
= 6

int *snitch
= ?

int bludger
? ?

Quidditch *
potter =

Stack (hufflepuff):

Hogwarts *
cedric

Quidditch *
seeker

int wizard =
3

Quidditch harry

int quaffle
= 8

int *snitch
=

int bludger
? ?

Quidditch *
potter = NULL

int = 2

int wizard =
3

Quidditch harry

int quaffle
= ?

int *snitch
= ?

int bludger
? ?

Quidditch *
potter = NULL

int wizard =
?

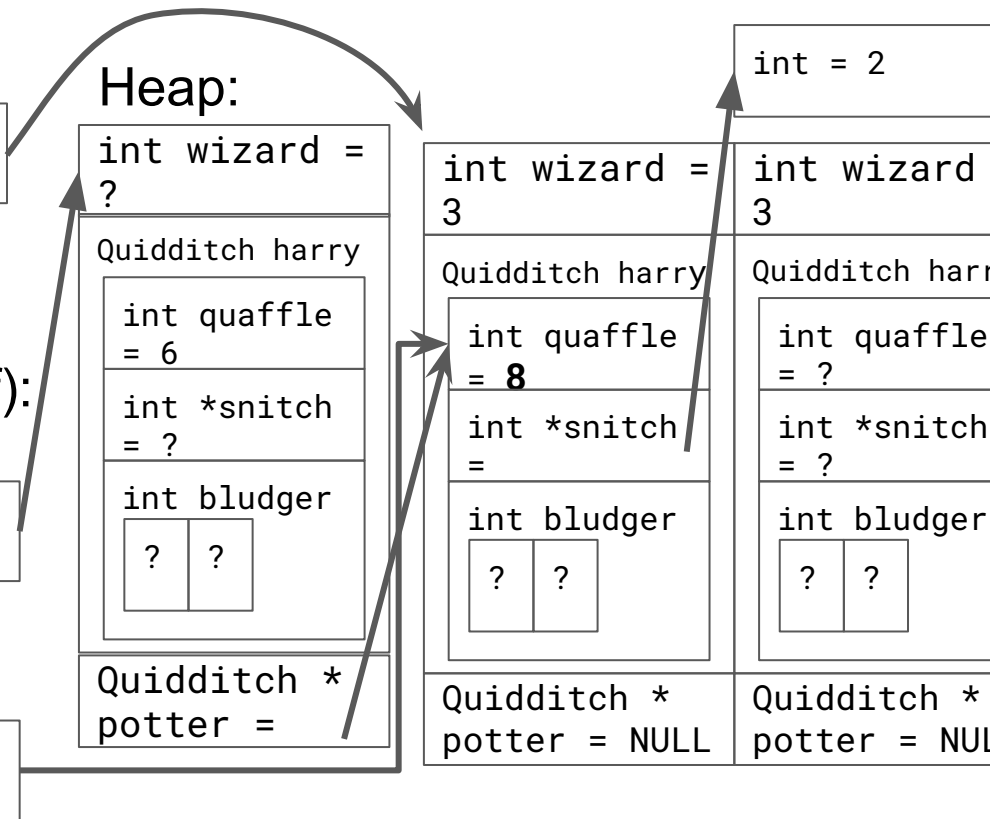
Quidditch harry

int quaffle
= ?

int *snitch
= ?

int bludger
? ?

Quidditch *
potter = ?



Pointer trace example

```
cedric->potter->snitch = &(cedric->potter->bludger[1]);
```

Stack (gryffindor):

Hogwarts *
triwizard

Heap:

int wizard =
?

Quidditch harry

int quaffle
= 6

int *snitch
= ?

int bludger
? ?

Quidditch *
potter =

Stack (hufflepuff):

Hogwarts *
cedric

Quidditch *
seeker

int = 2

int wizard =
3

Quidditch harry

int quaffle
= 8

int *snitch
=

int bludger
? ?

Quidditch *
potter = NULL

int wizard =
3

Quidditch harry

int quaffle
= ?

int *snitch
= ?

int bludger
? ?

Quidditch *
potter = NULL

int wizard =
?

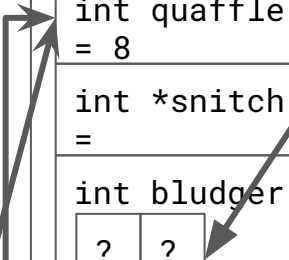
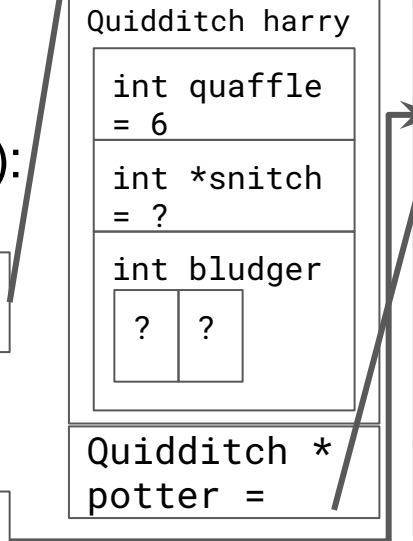
Quidditch harry

int quaffle
= ?

int *snitch
= ?

int bludger
? ?

Quidditch *
potter = ?



Pointer trace example

seeker->bludger[0] = 4;

Stack (gryffindor):

Hogwarts *
triwizard

Heap:

int wizard =
?

Quidditch harry

int quaffle
= 6

int *snitch
= ?

int bludger
? ?

Quidditch *
potter =

Stack (hufflepuff):

Hogwarts *
cedric

Quidditch *
seeker

int = 2

int wizard =
3

Quidditch harry

int quaffle
= 8

int *snitch
=

int bludger
4 ?

Quidditch *
potter = NULL

int wizard =
3

Quidditch harry

int quaffle
= ?

int *snitch
= ?

int bludger
? ?

Quidditch *
potter = NULL

int wizard =
?

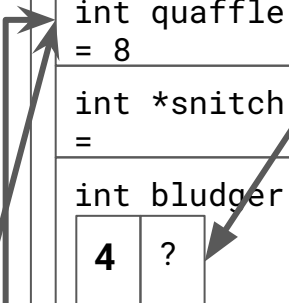
Quidditch harry

int quaffle
= ?

int *snitch
= ?

int bludger
? ?

Quidditch *
potter = ?



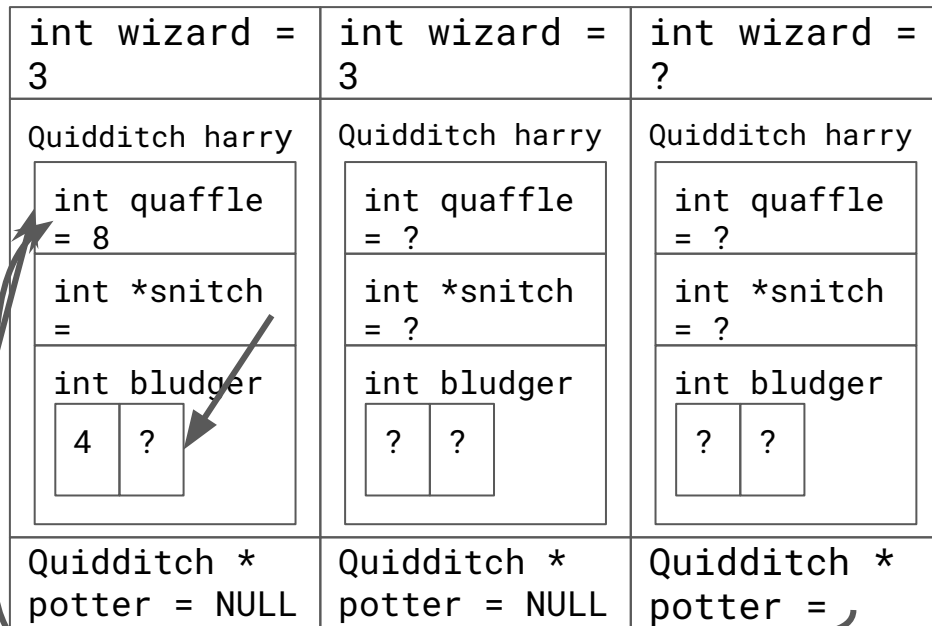
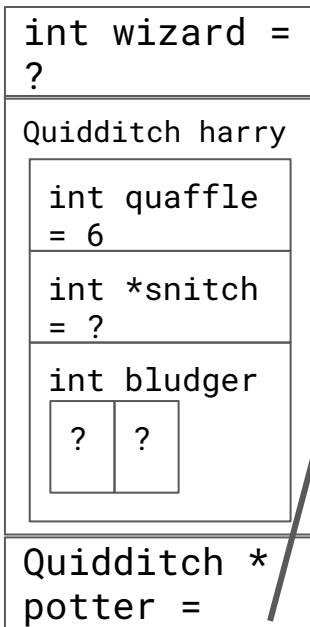
Pointer trace example

```
return seeker; triwizard[2].potter = hufflepuff(triwizard);
```

Stack (gryffindor):

Hogwarts *
triwizard

Heap:



Pointer trace example

triwizard[2].potter->quaffle = 4;

Stack (gryffindor):

Hogwarts *
triwizard

Heap:

int wizard = ?						
Quidditch harry						
<table><tr><td>int quaffle = 6</td></tr><tr><td>int *snitch = ?</td></tr><tr><td>int bludger</td></tr><tr><td><table><tr><td>?</td><td>?</td></tr></table></td></tr></table>	int quaffle = 6	int *snitch = ?	int bludger	<table><tr><td>?</td><td>?</td></tr></table>	?	?
int quaffle = 6						
int *snitch = ?						
int bludger						
<table><tr><td>?</td><td>?</td></tr></table>	?	?				
?	?					
Quidditch * potter =						

int = 2

int wizard = 3	int wizard = 3	int wizard = ?
Quidditch harry	Quidditch harry	Quidditch harry
int quaffle = 4 int *snitch = int bludger 4 ?	int quaffle = ? int *snitch = ? int bludger ? ?	int quaffle = ? int *snitch = ? int bludger ? ?
Quidditch * potter = NULL	Quidditch * potter = NULL	Quidditch * potter =

